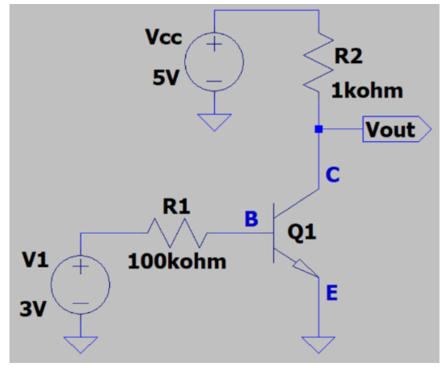
<u>Bipolar Junction Calculation HW (18 Points)</u>

(jas, BJT Calculation HW.docx, 2/26/2024)

1. Determine the base current I_B of Q1 in the adjacent circuit, given $V_{BE} = 0.7$ V. (Hint: Your answer should be between 20 and 30 µA.) (1 point.)

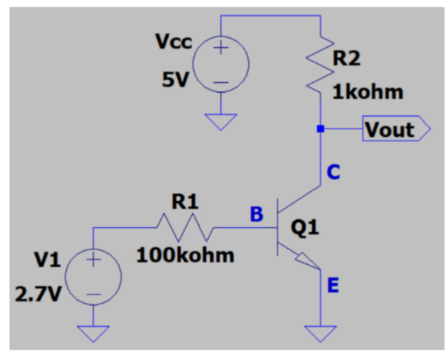


$$I_B = \frac{3 - 0.7\ \text{V}}{100 * 10^3 \Omega} = 23\ \mu\text{A}$$

$I_B = 23\ \mu\text{A}$

v

2. Determine the collector current I_C of Q1 in the adjacent circuit given $V_{BE} = 0.7$ V and $\beta = 100$. (Hint: Your answer should be between 1.8 and 2.2 mA.) (2 points.)

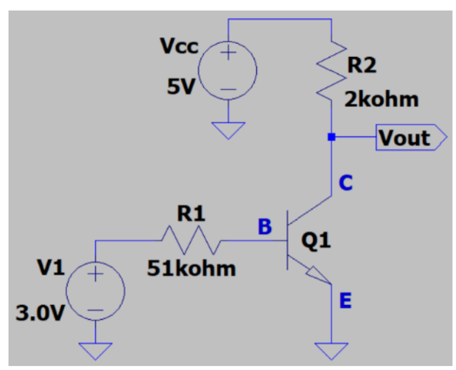


$$I_B = \frac{2.7 - 0.7\ \text{V}}{100 * 10^3 \Omega} = 20\ \mu\text{A}$$

$I_C = \beta I_B = 100(20\ \mu\text{A}) = 2000\ \mu\text{A} = 2.0\ \text{mA}$

$I_C = 2.0\ \text{mA}$

3. Determine the following for the adjacent circuit given $\beta = 53$ for Q1. (Hint: I_B should be between 40 and 50 µA.) (3 points total.)



   a. V_{out}. (2 points.)

$$I_B = \frac{3 - 0.7\ \text{V}}{51 * 10^3 \Omega} = 45\ \mu\text{A}$$

$I_C = 53(45\ \mu\text{A}) = 2.39\ \text{mA}$

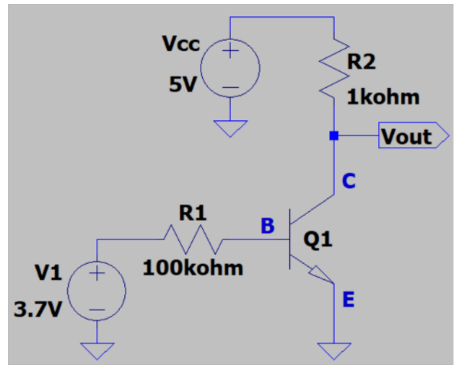
$V_{CE} = V_o = V_{CC} - V_{R2} = 5V - 2.39\ mA(2k\Omega) = 4.78\ V$

$V_{out} = 4.78\ \text{V}$

   b. The region of operation for Q1, with the choices being Cutoff, Circuit Saturation, or the Active Region. (1 point.)

$V_{CE} > 0.3\text{V} \rightarrow$ active region

4. Determine the following for the adjacent circuit given $\beta = 100$ for Q1. (Hint: I_B should be between 25 and 35 μA.) (3 points total.)



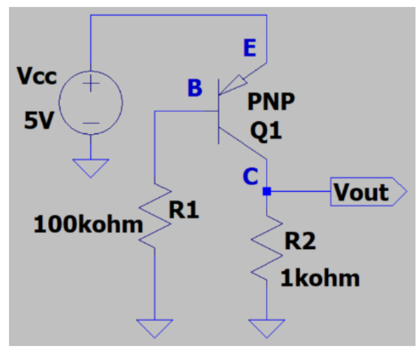
a. V_{out}. (2 points.)

$$I_B = \frac{3.7 - 0.7\ \text{V}}{100 * 10^3\Omega} = 30\ \mu\text{A}$$
$$I_C = 100(30\ \mu\text{A}) = 3.0\ \text{mA}$$
$$V_{R2} = 3\text{mA} * 1\text{k}\Omega$$
$$V_{BE} = V_{out} = V_{CC} - V_{R2} = 2\ \text{V}$$
$$V_{out} = 2\ \text{V}$$

b. The region of operation for Q1, with the choices being Cutoff, Circuit Saturation, or the Active Region. (1 point.)

$V_{BE} = 2\ \text{V} < 0.7\ \text{V} \rightarrow$ Active

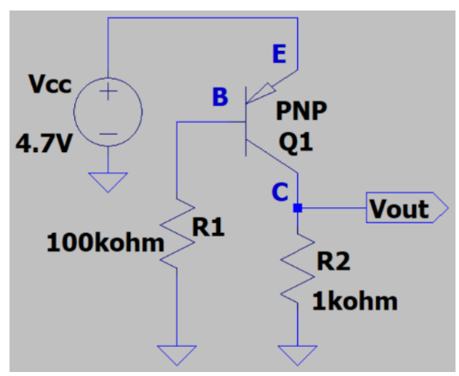
5. Determine the base current I_B of Q1 in the adjacent circuit given $V_{EB} = 0.7$ V and noting that conventional current is considered negative when flowing out of a transistor terminal. (Hint: Your answer should have a magnitude between 40 and 50 μA.) (1 point.)



$$I_B = \frac{V_{CC} - V_{EB}}{R1} = \frac{5 - 0.7}{100\ \text{k}\Omega} = 43\ \mu\text{A}$$

$I_B = -43\ \mu\text{A}$ (flowing out of base)

6. Determine the collector current I_C of Q1 in the adjacent circuit given $V_{BE} = 0.7$ V, $\beta = 110$, and noting that conventional current is considered negative when flowing out of a transistor terminal. (Hint: Your answer should have a magnitude between 4.2 and 4.6 mA.) (2 points.)

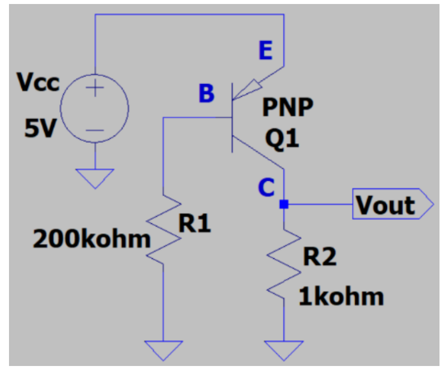


$$I_B = \frac{V_{CC} - V_{BE}}{R1} = \frac{4.7 - 0.7\ \text{V}}{100 * 10^3}\Omega = 40\ \mu\text{A}$$

$$I_C = \beta I_B = 110(40\ \mu\text{A}) = 4.4\ \text{mA}$$

$I_C = 4.4\ \text{mA}$

7. Determine the following for the adjacent circuit given $\beta = 95$ for Q1. (Hint: $|I_B|$ should be between 18 and 25 µA.) (3 points total.)



a. V_{out}. (2 points.)

$V_{BE} = 0.7\text{ V}$

$I_B = \dfrac{5 - 0.7\text{ V}}{200 * 10^3\ \Omega} = 21.5\ \mu\text{A}$

$I_C = \beta I_B = 95(21.5\ \mu\text{A}) = 2.04\text{ mA}$

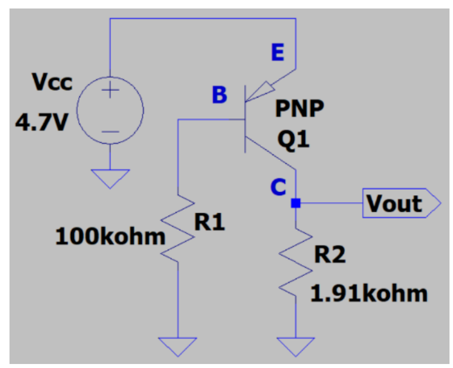
$V_o = I_C(R_2) = 2.04\text{ mA}(1\text{ k}\Omega) = 2.04\text{ V}$

$V_o = 2.04\text{ V}$

b. The region of operation for Q1, with the choices being Cutoff, Circuit Saturation, or the Active Region. (1 point.)

$V_{CE} = V_{CC} - V_O = 5\text{V} - 2.04\text{V} = 2.96\text{V} \rightarrow$ active region

8. Determine the following for the adjacent circuit given $\beta = 60$ for Q1. (Hint: $|I_B|$ should be between 30 and 50 µA.) (3 points total.)



a. V_{out}. (2 points.)

$V_{BE} = 0.7\ V$

$I_B = \dfrac{4.7 - 0.7\ V}{100\ k\Omega} = 40\ \mu A$

$I_C = \beta I_B = 60(40\ \mu A) = 2.4\ mA$

$V_o = I_C(R_2) - 2.3\ mA\ (1.91\ k\Omega) = 4.58\ V$

$V_o = 4.58\ V$

b. The region of operation for Q1, with the choices being Cutoff, Circuit Saturation, or the Active Region. (1 point.)

$V_{CE} = 5\text{V} - 4.58\text{ V} = 0.416\text{ V} > 0.3 \rightarrow$ active